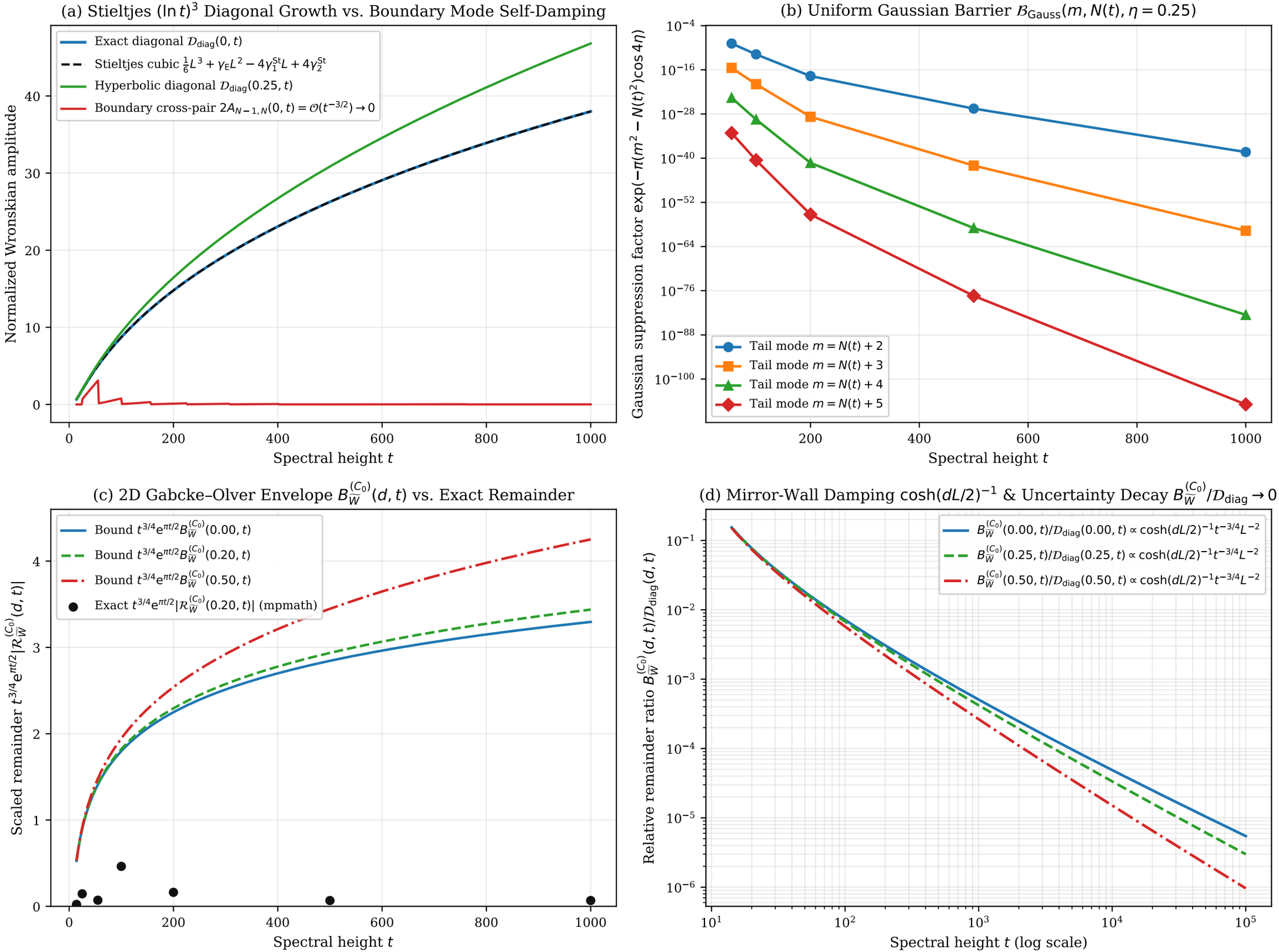

(a) Stieltjes $(\ln t)^3$ Diagonal Growth vs. Boundary Mode Self-Damping
Exact diagonal $\mathcal{D}_{\text{diag}}(0,t)$
Stieltjes cubic $\frac{1}{6}L^3+\gamma_E L^2-4\gamma_1^{\text{St}}L+4\gamma_2^{\text{St}}$
Hyperbolic diagonal $\mathcal{D}_{\text{diag}}(0.25,t)$
Boundary cross-pair $2A_{N-1,N}(0,t)=\mathcal{O}(t^{-3/2})\to 0$
Normalized Wronskian amplitude
Spectral height t
(b) Uniform Gaussian Barrier $\mathcal{B}_{\text{Gauss}}(m,N(t),\eta=0.25)$
Gaussian suppression factor $\exp(-\pi(m^2-N(t)^2)\cos 4\eta)$
Tail mode $m=N(t)+2$
Tail mode $m=N(t)+3$
Tail mode $m=N(t)+4$
Tail mode $m=N(t)+5$
Spectral height t
(c) 2D Gabcke–Olver Envelope $B_{\widetilde{W}}^{(C_0)}(d,t)$ vs. Exact Remainder
Bound $t^{3/4}e^{\pi t/2}B_{\widetilde{W}}^{(C_0)}(0.00,t)$
Bound $t^{3/4}e^{\pi t/2}B_{\widetilde{W}}^{(C_0)}(0.20,t)$
Bound $t^{3/4}e^{\pi t/2}B_{\widetilde{W}}^{(C_0)}(0.50,t)$
Exact $t^{3/4}e^{\pi t/2}|\mathcal{R}_{\widetilde{W}}^{(C_0)}(0.20,t)|$ (mpmath)
Scaled remainder $t^{3/4}e^{\pi t/2}|\mathcal{R}_{\widetilde{W}}^{(C_0)}(d,t)|$
Spectral height t
(d) Mirror-Wall Damping $\cosh(dL/2)^{-1}$ & Uncertainty Decay $B_{\widetilde{W}}^{(C_0)}/\mathcal{D}_{\text{diag}}\to 0$
$B_{\widetilde{W}}^{(C_0)}(0.00,t)/\mathcal{D}_{\text{diag}}(0.00,t)\propto\cosh(dL/2)^{-1}t^{-3/4}L^{-2}$
$B_{\widetilde{W}}^{(C_0)}(0.25,t)/\mathcal{D}_{\text{diag}}(0.25,t)\propto\cosh(dL/2)^{-1}t^{-3/4}L^{-2}$
$B_{\widetilde{W}}^{(C_0)}(0.50,t)/\mathcal{D}_{\text{diag}}(0.50,t)\propto\cosh(dL/2)^{-1}t^{-3/4}L^{-2}$
Relative remainder ratio $B_{\widetilde{W}}^{(C_0)}(d,t)/\mathcal{D}_{\text{diag}}(d,t)$
Spectral height t (log scale)